

Checkable Properties for a Research Record: A Specification, a Reference Validator, and Three Defects It Found That Careful Reading Did Not

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Abstract—Research integrity rests on claims a reader is asked to take on trust. A paper asserts that its hypotheses preceded its measurements, that nothing was omitted, and that its conclusions do not outrun its evidence, and none of these assertions is checkable from the artifact that makes them. We specify four properties of a version-controlled research record that convert those assertions into predicates a machine can refuse, implement a reference validator, and run it against two live corpora. The validator found three classes of defect that accurate prose had absorbed silently. Ordinary version-control hygiene destroyed the evidence for twelve priority claims without anyone noticing, which forced a change to the specification. A correction was propagated so completely that the published book reads as though the stronger claim had never been made, leaving a blast radius of zero that is the finding rather than the absence of one. Three results were graded as requiring no modeling assumptions while their own dependency lists named them. The validator also returned a result against the specification that defines it: it was certifying the coherence level for a deployment whose dependency graph its parser cannot populate, so that check was deciding an empty graph. The same programme requires judgment procedures to prove they are not vacuous, and the record specification carried no equivalent condition. It now does, and the reference deployment moves down a level as a result. What it demonstrated was never coherence, only the absence of anything to be incoherent about.

Index Terms—Research integrity, reproducibility, preregistration, provenance, scientific record, audit, accountability, research software.

I. INTRODUCTION

A RESEARCH paper makes claims about the world and claims about itself. The first kind is scrutinized. The second kind is not, because it cannot be. When a paper states that its hypotheses were fixed before the data were seen, that no analyses were run and discarded, and that its conclusions are supported by the evidence cited, a reader has no way to test any of it from the artifact in front of them. These are assertions about an unobservable process, made by the only party with an interest in the answer.

Reform has concentrated on external registries, which work and which we build on. Preregistration establishes priority through a trusted third party. Registered reports make it a condition of publication. Both operate on the study as a unit.

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Specification, validator, and both conformance reports are public at <https://github.com/ahb-sjsu/philosophy-engineering>.

Neither addresses what happens afterwards, inside a body of work, when one result fails and the author must determine what else falls.

Four difficulties survive that reform, and all four are structural rather than cultural. The unit of publication is the paper while the unit of truth is the claim, so a paper with nine sound results and one bad lemma has no way to express that state. Because the artifact is indivisible, correcting the lemma means retracting everything bundled with it, which makes correcting nothing the rational choice. Nothing records what depended on the lemma, so the question of what else falls is answered from memory. And the absence of a result is invisible by construction, which makes any claim to have reported everything a claim about a set no one can inspect.

This paper reports an attempt to make those four assertions checkable. We specify four properties of a research record held in version control, implement a validator that decides them, and run it against two corpora that were not built for it. Our contribution is not the components. Claim-level publication, argumentation graphs over evidence, and version-controlled manuscripts all predate this work by a decade or more, and Section IX says so at length. The contribution is the conjunction, operated against live corpora, and the measurements that follow.

Three results are worth stating in advance. First, the checks found defects that careful reading had not, and the defects were of a kind prose cannot hold rather than a kind the authors were careless about. In one corpus, ordinary rebasing destroyed the evidence for twelve priority claims while the written record remained accurate throughout. Second, one class of accountability failure appears to be undetectable by any existing mechanism: a correction can be propagated so thoroughly that no trace of the retired claim survives, which leaves a published work reading as though it had never erred. We recover it only because an append-only record holds the claim and its withdrawal at the same time. Third, running the validator against our own specification exposed a gap in the specification. It certifies the coherence level for a deployment whose dependency graph is empty, because the parser for that deployment's format cannot represent dependency edges. A sibling part of the same programme requires judgment procedures to demonstrate that they are not vacuously compliant, and the record specification has no matching requirement.

TABLE I
THE FOUR PROPERTIES. EACH CONVERTS AN ASSERTION NORMALLY
TAKEN ON TRUST INTO A PREDICATE OVER A VERSION-CONTROLLED
RECORD.

	The assertion	The predicate
P1	We registered before we measured	The sealing commit is a git ancestor of the first commit adding the result
P2	There is no file drawer	The identifier sequence is gap-free and every identifier resolves to a disposition
P3	We do not assert beyond our evidence	No narrative document cites a claim above its class
P4	We know what depends on what	Every declared dependency resolves, and no claim outranks its weakest support

II. FOUR PROPERTIES

The record is a set of claim objects in a git repository. Each carries an identifier, a statement, an evidence class, declared dependencies on other claims, and where applicable a registration sealed before measurement. Classes are ordered by the support they assert, from *exploratory* through *demonstrated* and *replicated* to *proved*, with *withdrawn* and *refuted* for claims that have lost support. The four properties are predicates over that structure.

P1 uses repository ancestry rather than a timestamp.

A third-party registry proves wall-clock time and requires trusting the registry. `git merge-base --is-ancestor` proves order and requires trusting nothing, but is checkable by anyone holding a copy of the repository. Section IV reports what this choice costs.

P2 is the property we would defend first. No file drawer is ordinarily an unfalsifiable claim, since it quantifies over work that was never recorded. Requiring identifiers to be assigned contiguously, and requiring every identifier to carry a disposition including the abandoned and the never-run, converts it into a property of a finite list. A programme hiding a failed run under this regime must either leave a visible gap or write a false disposition. The guarantee is bounded by where the identifier space begins, which Section VIII treats as the limitation it is.

P4 makes the consequences of a refutation computable.

When a claim fails, the set of claims resting on it is obtained by traversing declared edges rather than by recollection. We call the result the blast radius. The design intent is that a bad result prunes a branch instead of burning the tree, and Section V measures whether it does.

Conformance is layered. L1 requires the objects and P2. L2 adds P1. L3 adds declared dependency edges, P4, P3, and blast-radius reporting. L4 additionally requires fresh-context verification records and machine-checked cores as linked objects. L1 is achievable by one researcher in an afternoon and already delivers the file-drawer guarantee.

III. THE VALIDATOR

`pe_lint.py` is 939 lines of Python with no dependencies outside the standard library and `git`. It parses two record formats, decides P1 through P4, computes blast radius, and

exits non-zero when the requested level is not met, which makes it usable as a continuous-integration gate.

Two design choices are worth reporting because both were forced by deployment.

The ratchet. A conversion of an existing corpus surfaces contradictions that the converter is forbidden to resolve, because resolving them would mean rewriting the work rather than describing it. Left alone, these hold the gate permanently red, and a gate that always fails gates nothing. The validator therefore accepts a baseline of acknowledged findings, each naming an owner and what would settle it. The mechanism is built so that it cannot be used to launder claims. Entries match on exact finding text, so a promoted claim stops matching and reappears. A stale entry fails the run. A missing owner fails the run. No run with a non-empty baseline may report unconditional conformance, and the report says so on its own output. A baseline may only shrink.

Two front ends. One deployment stores claims as native JSON objects. The other predates the specification and stores them as markdown tables. The markdown parser recovers identifiers, classes, statements, and registrations. It does not recover dependency edges, because the format has no field for them. Section VI reports what that costs, and it is the most serious finding in this paper.

IV. DEPLOYMENT A: A RECORD BUILT UNDER THE SPECIFICATION

The first deployment is the evidence ledger of an observation-theory programme: 78 claims, and a registry of 91 sealed registrations.

Under the amended specification of Section `efsec:selfcheck` it conforms at L2, and the L3 it was previously reported at was vacuous. P2 holds over the full range 001 to 091 with zero gaps and four identifiers carrying no result: two never assigned and proven absent from the full history, one reservation abandoned before sealing, and one burned by a design failure that its own pilot caught. P4 reports no support-cap violations. P3 scans three narrative documents and raises two warnings. Priority verifies on all 32 seals that carry checkable anchors.

That clean report is the second one. The first run failed, and the failure is the useful part.

Twelve priority proofs had been destroyed by ordinary tooling. One seal resolved but failed ancestry. Eleven named commits that no longer existed. The cause was the same in both cases and it was not misconduct: rebasing rewrites commit hashes, and a recorded seal hash is a reference to an object that rebasing destroys. The eleven were sealed during a period when the author was pushing from several machines with rebase-on-pull. The priority was genuine in every case. The evidence for it was collateral damage from routine version-control hygiene, and the registry's own notes show the author had observed the rebase without observing that it had broken the proof.

This forced a normative addition to the specification. A conforming record must not rewrite history containing seal commits. Registrations must carry a content hash independent

of commit identity, so that a rewritten record can be re-anchored. A rewritten seal must record the old-to-new mapping in its row. And a non-resolving seal without such a mapping is a failure rather than a warning, because a claim whose priority cannot be checked is a claim whose priority is unproven.

Remediation recovered all twelve. The re-anchoring script refuses to write unless the content hash still matches under the repository's own sealing scheme, the commit is the first that added the registration file, and it is an ancestor of the result commit. A pre-rebase hook now refuses any rebase touching the registration directory. A separate script anchors the content hash to a third-party timestamp, which closes the gap that ancestry leaves open: ancestry proves order, and an anchor proves wall-clock time to a stranger.

Remediation then found a third defect. Two registrations failed the content-hash check. Both carried dated, disclosed amendments appended to the sealed body after sealing, which the protocol permits. Appending them changed the body, so the seal hash stopped verifying. The specification had required amendments to be dated and reasoned and had said nothing about their effect on the content commitment. The correct design, now stated, is that the sealed body is immutable and amendments are appended as separately hashed records.

None of these three defects was detectable by reading. The written record was accurate throughout.

V. DEPLOYMENT B: CONVERTING AN EXISTING CORPUS

The second deployment is a book of roughly thirty-five thousand lines with associated papers, written over several years before the specification existed. Conversion produced 236 claim objects. It is descriptive first: no claim was re-derived, no statement was edited, and nothing was deleted, on the ground that priority is the one property that no later rigor reconstructs.

The fraction of claims entered without a prior registration is 0.958. That number is a measurement rather than a criticism, and near unity is expected by construction for any corpus converted after the fact. What matters is its trajectory.

A. The blast-radius distribution

The corpus declares 196 dependency edges, with 154 of 236 claims carrying at least one. Fig. 1 shows the blast radius of every claim, computed by traversal.

Mean blast radius is 2.77 and the median is 0. The maximum in-degree is 33. The shape is the one a theory of research programmes would predict, in which refutations ordinarily strike auxiliary claims and leave a protected core untouched, and the distribution locates that core in a particular place: the seven claims with the largest blast radius are all definitions of the underlying mathematical objects.

We flag an ambiguity rather than resolve it. A heavy tail concentrated on a few high in-degree claims is consistent with a genuine foundational core, and it is equally consistent with dependencies being under-declared everywhere else. This corpus gives evidence for both readings. The conversion logged 29 claims whose declared dependencies point outside the

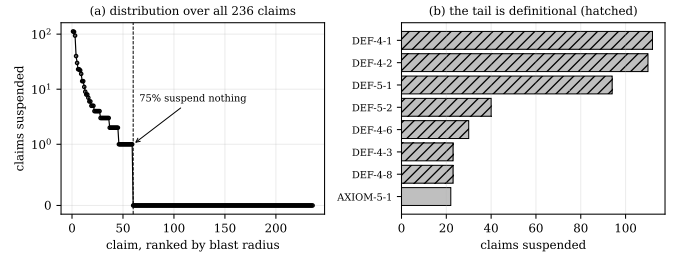


Fig. 1. Blast radius of every claim in Deployment B, computed from the live record. (a) The distribution is sharply right-skewed: 75.0% of claims suspend nothing at all and 84.8% suspend two or fewer, while the largest suspends 112. (b) The tail is definitional. The claims whose refutation would reach furthest are the definitions of the underlying mathematical objects, not the empirical results. A refutation in this corpus almost always prunes a branch; the exceptions are foundational by construction.

record entirely, and wired only 49 of 81 previously untagged results to their roots. The distribution in Fig. 1 is therefore measured on a graph its own conversion report describes as incomplete, and we draw no conclusion about research programmes in general from a single partial graph.

B. Two findings the checker made visible

A refutation that pruned five claims out of 140. The corpus asserts a symmetry group larger than the one its evidence supports. The measured result supports a four-element group; a larger group is posited and would require an operation that is nowhere measured. Suspending the over-strong claim suspends five dependent theorems and leaves the rest untouched, and the report names both sets explicitly. That is the intended behavior, and it is the difference between pruning and burning.

A correction with a blast radius of zero, which is the finding. The corpus had committed to a conservation law as a chapter thesis. The author later judged the claim to overreach and retreated to a weaker statement about invariance, giving the reason in a commit message. The work was careful: 87 occurrences rewritten, text nodes only, with all 2,126 equations verified identical before and after. It was also complete. The current version contains zero occurrences of the retired phrasing and 47 of the replacement.

That completeness is the problem. The published book now reads as though the stronger claim was never made, and a reader cannot discover that a thesis was retired, because every trace of it was correctly removed. The only surviving record of the retreat was a commit message. Entered into the record as *withdrawn*, the claim reports a blast radius of zero, and the zero is the result: it is the formal statement that this correction was propagated perfectly and recorded nowhere.

We think this is the most consequential finding in the paper for research accountability, because no existing mechanism detects it. Retraction notices address papers that are withdrawn. Version control records that text changed. Neither surfaces a thesis that was quietly weakened and then made invisible by a thorough edit. A programme that counted only suspensions would score this episode as a non-event. The general statement is uncomfortable: a retraction thorough enough to be invisible is indistinguishable from never having erred, and only an

append-only record holds the claim and its withdrawal at the same time.

C. Contradictions only the author can settle

Three results are graded as following from standard mathematics without modeling assumptions, while their own dependency lists name modeling constructs. A result cannot be both. Either the grade or the dependency list is wrong, and the specification forbids the converter from deciding which. These are the record's four surviving coherence errors, they are acknowledged in the baseline with a named owner, and the current run reports conformance against that baseline while stating on its own output that it is not unconditional. Four further claims raise warnings for being classified as definitions while resting on substantive claims, which the specification treats as a modeling assumption in disguise.

The conversion also found 81 numbered results carrying no epistemic grade and appearing in no registry, roughly a third of the corpus's formal content, and therefore invisible to the corpus's own accounting before conversion.

VI. WHAT THE VALIDATOR FOUND IN THE SPECIFICATION

The result with the most consequence here came from pointing the validator at the specification that defines it. We report it the way we report the others, because a check that returns something is a check doing its work, and a specification that could not be caught by its own instrument would be the more troubling object.

Deployment A is certified at L3. L3 requires declared dependency edges, P4, and blast-radius reporting. Deployment A's record is stored as markdown tables, and the markdown parser assigns no dependency edges, because the format has no field for them. We verified this directly: zero of its 78 claims carry a dependency edge, and the parsing routine never assigns the field. P4 therefore evaluates over an empty graph, finds no violations because there is nothing to violate, and reports PASS. Blast radius is zero for every claim in that deployment for the same reason. The validator reports conformance at the level whose entire purpose is dependency structure, for a record that declares none.

This is a vacuous-compliance failure, and it is a failure the same programme already knows how to prevent. Elsewhere in this body of work, a judgment procedure is required to be invariant under declared transformations, and that requirement is paired with an explicit non-degeneracy condition, because a constant function satisfies invariance perfectly and is useless. The record specification imposes no analogous condition on its own dependency graph. It demands non-degeneracy of the systems it governs and does not demand it of itself.

The fix is straightforward to state. L3 should require dependency declaration to be non-vacuous, by reporting edge coverage as a first-class statistic and refusing to certify blast-radius conformance when the graph is empty or when coverage falls below a declared threshold. A format that cannot express dependency edges should be capped at L2 rather than certified at L3.

We implemented it. The specification gained a non-vacuity clause requiring that edge coverage be reported in every run, that a ledger declaring no resolving dependency edge fail, and that a storage format incapable of expressing such an edge be capped at L2, where no property quantifies over the graph. A programme may declare a minimum coverage and the validator enforces it; the specification declines to set a general floor, because the defensible value depends on how many of a corpus's claims are independent of the rest.

The effect on the two deployments is asymmetric. Deployment A now fails L3 and conforms at L2. Nothing about that programme changed: its claims, its seals, and its registry accounting are exactly as previously reported, and no scientific result is affected. What changed is that the report no longer states that coherence was verified when what was verified was an empty graph. Deployment B is unaffected, declaring 196 resolving edges across 154 of its 236 claims, an edge coverage of 0.653, and continuing to conform at L3 against its baseline.

The gap had persisted through a written conformance report and a published case study without a reader noticing, which is the same pattern as the other three findings and the reason we think the exercise is worth the cost. Readers checked whether the coherence property passed. Deciding whether it had anything to pass over is the kind of question a check answers cheaply and a reading does not answer at all.

VII. DISCUSSION

The defects reported here fall into three classes, and the classes suggest what this kind of checking is for.

The first class is identity loss under ordinary tooling. Twelve priority proofs were destroyed by rebasing, in a repository whose written record was accurate throughout, and no reader could have found it. This class is invisible to human review because the thing that broke was a machine-checkable reference and not a statement.

The second class is the complete and unrecorded correction. It is invisible for the opposite reason: the work was done properly and the evidence of it was removed as part of doing it properly. Only an append-only structure catches this, and catching it requires the counterintuitive step of treating a blast radius of zero as a result rather than as silence.

The third class is self-contradiction across distance. A result graded as assumption-free while its own dependency list names assumptions, or a falsification condition contradicted 112 lines later in the same file, is a defect a reader can find only by holding two distant passages in mind at once. Machines hold distant passages in mind cheaply.

What checking does not do is also worth stating. It cannot tell whether a claim is true. It cannot propagate along dependencies nobody declared, which is the classical point that no hypothesis is tested in isolation and the reason undeclared edges are this method's blind spot. It cannot decide the two P3 warnings in Deployment A, because distinguishing a citation the claim rests on from a contextual one requires knowing which the author intended, and the markdown front end does not record it.

There is a governance problem that the technical work does not touch. Every institution that made record-keeping stick

had an outside party entitled to inspect and to impose a cost: journals enforce registration, regulators enforce trial registries, editors enforce review. A record operated by the only party it constrains has no such enforcement, and the properties it checks are adopted voluntarily by the person they restrain. Nothing in the specification fixes this, and we think it is the most likely way for the method to fail in practice while appearing to succeed on paper.

VIII. LIMITATIONS AND DISCLOSED NEGATIVES

Two corpora, both maintained by this paper's author, who chose the method and then reported that it worked. The confirmation risk is severe and cannot be reduced from inside. Nothing here has been tried by a programme with different authorship, and the measurements in Fig. 1 should be read as one observation rather than as a distribution over research programmes.

A companion set of checks, governing which transformations a claim was tested against, does not pass on this programme. Ten tests cite three transformation families that the relevant registry never declares. The families were declared in the individual sealed registrations and were not mirrored into the registry.

That finding produced the second specification change reported in this paper, by the same route as the first. The specification admitted only a conforming row or a missing one, so a registry repaired after the fact was indistinguishable from a registry whose families were chosen to fit results already seen. It now distinguishes them. A late row is admissible as a backfill when it carries the date it was written, names the sealed artifact holding the real declaration with path, content hash, and sealing commit, and when that declaration demonstrably precedes the result, which means the seal commit is an ancestor of the commit adding the result and the sealed artifact still hashes to its recorded value. Where the third condition cannot be shown the row must not be entered, and deleting the offending tests is not a repair either, because removing the record of a test that ran is the file drawer P2 exists to prevent. Backfilled rows are reported rather than passed over, on the reasoning that a registry with many of them belongs to a programme that is not writing declarations down when it makes them.

We do not report whether the reference registry has since been repaired under that clause. Those registries are not public, and this paper verifies only what it can run.

P2's guarantee is narrower than its name. Contiguous identifiers make the set of identified lines of work enumerable. They say nothing about work considered and never given an identifier, and the boundary of the identifier space is set by the same party the property is meant to constrain.

Deployment A does not claim L4. Fresh-context verification records and machine-checked cores both exist in that programme, but they are recorded in prose rather than as linked objects, and claiming a level whose requirement is linked objects would be the error this apparatus exists to prevent.

Finally, Section VI's clause leaves the two deployments incomparable on coherence, and now says so rather than

obscuring it. Deployment B is evaluated over a real dependency graph at 0.653 edge coverage. Deployment A has no graph to evaluate, and recovering one means migrating it to a format with a dependency field and declaring structure that currently exists only implicitly in its chapters. That migration is unperformed, and until it is done this paper reports one deployment's coherence rather than two.

IX. RELATED WORK

Preregistration establishes the norm P1 enforces, and registered reports make it a publication condition, at a scale this work does not approach. The differences are that registries are external services whose timestamps must be trusted, and that registration is per-study rather than per-claim with a dependency graph.

Claim-level publication was proposed as nanopublications, in which the smallest publishable unit is a single assertion with its provenance, and developed as micropublications, which link claims to evidence and to each other with support and challenge edges. Both predate this specification and both have better formal serializations. They solve granularity and linkage. Neither makes the absence of a claim observable, neither establishes that a prediction preceded its test, and neither defines what a challenge does to the class of a dependent claim.

Manubot treats manuscripts as version-controlled repositories with continuous integration, which is the infrastructure model assumed here; it versions the document, and this work versions the claims inside it. Proof-assistant libraries already compute blast radius exactly, for the fragment of claims that can be formalized, and are the closest existing system to the ideal. The contribution here is an attempt to extend that discipline to empirical claims, where no compiler exists and edges must be declared by hand.

Lakat proposes a peer-to-peer architecture for continuous-integration publishing with a review-based consensus mechanism. Its concern is the governance of a distributed publishing system rather than the appraisal of a single programme's record. The broader reform movement in psychology reached the conclusion that changes of this kind are reform of evidentiary standards rather than changes of theory, and we agree.

We do not offer a better registry, a better claim format, or a better proof assistant. We offer the conjunction, and the measurement of what it caught.

X. CONCLUSION

Four assertions that research artifacts normally make on trust can be stated as predicates over a version-controlled record, and a 939-line checker can decide them. Run against two corpora that were not built for it, the checker found priority evidence destroyed by routine tooling, a correction so complete that it erased the record of its own occurrence, and results graded above the support their own dependency lists declare. None of the three was visible to careful reading, and the written record was accurate in every case.

The checker also found that our specification certifies its coherence level for a record with no dependency graph, which

is the vacuous-compliance failure that the same programme requires the systems it governs to rule out. We have stated the fix and left it unapplied, because it changes the standing of a deployed record and that decision belongs to the specification's owner.

The property we would defend first is the second one. No file drawer is normally a claim about a set nobody can inspect. Assign identifiers contiguously, require every identifier to carry a disposition including the abandoned and the never-run, and it becomes a property of a finite list that anyone can check in seconds. That conversion, from an unfalsifiable assurance to a finite check, is what the rest of this apparatus is for.

COMPANION PAPER

The question of why these four properties and not others is a question about the appraisal of research programmes, and it is treated separately in a companion manuscript that carries no empirical content. That paper argues that declaring a dependency partition before a test, and proving priority by repository ancestry, converts a distinction that philosophy of science could previously draw only in hindsight into a property of the record. The two papers are separable: this one reports what a checker found and stands without the argument, and the companion makes an argument and states explicitly that it has no measurements. The blast-radius distribution in Fig. 1 is the first test of a prediction that companion states.

ACKNOWLEDGMENT OF AI USE

Drafting, prior-art search, figure construction, and structural editing were performed with the assistance of a large language model under the author's direction. Every quantity reported here was produced by the scripts in the repository's `build` directory, reading the live records, rather than transcribed. The author is responsible for the claims and the verdicts.

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